

Interaction

Now we are going to discuss the 5 concepts that are the backbone of interaction design:

- 1) Affordance
- 2) Conceptual Model
- 3) Natural Mapping
- 4) Feedback
- 5) Constraints

1) Affordance:

"Does the user know what actions can be done?"

An **affordance** is a property of an object that shows what actions are possible. It determines how an object can be used.

day/date

e.g.

If a door has a handle, it shows pull.

• A flat plate on door shows push.

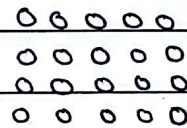
• A 3D styled button affords clicking

• A slider thumb affords dragging

We can use Gestalt principles to help create & reinforce perceived affordances

Gestalt Laws:

1) Proximity:

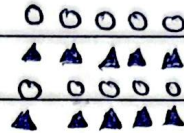


Elements positioned close to each other are perceived as a single group or unit.

Bereh: Group Interactive controls
together to signify unified function.

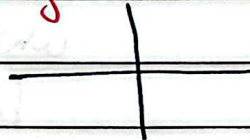
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2) Similarity:



• Elements that look similar are processed to share same function or meaning

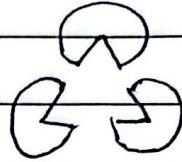
3) Continuity:



The human eye naturally follows a path, a line or curve.

eg A row of image cards in websites with one image partially cut off on screen ~~shows~~ affords swiping horizontally.

4) Closure:



The mind automatically fills in missing visual information to perceive a complete shape.

- Use minimal visual elements while maintaining clear functional affordances.

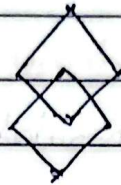
~~5) Figure / Ground:~~

5) Area:



Elements in same boundary are perceived as belonging together.

c) Symmetry



visual elements that are arranged symmetrically around a central axis are perceived as having equal importance.

2) Conceptual Model:

A **Conceptual Model** is a high level mental stimulation formed by the user about how a system works.

"Does the user understand how the system operates?"

E.g. The concept of Desktop, Folders & Trash can on an OS is a conceptual model designed to mimic a real physical office desk.

3) Natural Mapping:

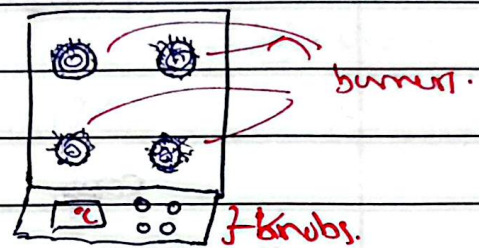
Mappings is the relationship between spatial controls & their results in the physical or digital world.

"Is the relationship between control & effect obvious?"

Example:

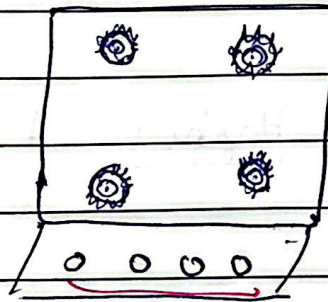
Good Mapping:

Arranging 4 stove-top knobs in the exact same spatial layout as the 4 burners.



Bad Mapping:

Arrange the 4 knobs in different layout



now user needs to guess which knob is of which burner

4) Feedback:

It is the sending information back to user about what actions has been performed.

Feedback must be: **immediate, clear & unambiguous.**

"Does user know what system did?"

Example:

visual: A button changing color when ~~click~~ pressed.

Auditory: A click sound when locking phone.

Haptic: A vibration when you tap on screen.

5) Constraints:

Constraints limits the set of possible actions a user can perform at any given moment

"Is the user restricted from making any critical errors"

Example:

- You can only insert an SD-card or USB-C cable in specific ways.

- Cultural constraints: Red = Stop, green = go.

A few evaluation metrics:

Now we will see a few evaluation metrics that can be used to check the interactivity & ease of use of anything:

1) Fitt's law:

It predicts the time required to rapidly move to a target area as a function of target's distance (D) & its width (w).

$$MT = a + b \cdot \log_2 \left(\frac{2D}{w} \right)$$

MT = movement Time

a, b = constants.

D = distance

w = width } of target.

· increasing w reduces movement time.

2) Hick Hyman's Law (Hick's Law)

This tells the ~~time to make~~ ^{time to make} ~~number of~~ ^{of} decisions when "n" number of choices are present.

$$T = b \cdot \log_2(n+1)$$

n = equal probability choices.

b = constant.

3) Error Rate:

$$\text{Error rate per task} = \frac{\text{Total errors across all users}}{\text{Total task Attempts}}$$

4) Task completion Rate.

$$\text{Completion Rate} = \frac{\text{Successfully completed tasks}}{\text{Total tasks}} \times 100$$

5) Throughput.

This measures the overall information processing capacity of a system.

$$TP = \frac{IDe}{MT}$$

MT = movement time from Fitts law.

$$IDe = \log_2 \left(\frac{D}{w_e} + 1 \right)$$

D = Distance

w_e = Effective width.

Traditional Interaction Techniques:

Now we will look at a few interaction techniques & choose the right one.

Which interaction technique is Best?

To answer this questions we need to see
3 things:

- 1) User : who is interacting
- 2) Task : what must be done
- 3) Context : where & under what conditions.

Best technique = The one that fits the
User + Task + Context.

day/date

User

Who is interacting?

- Novice or expert
- Motor abilities
- Preferences
- Cognitive workload

Task

What must be done?

- Text entry
- Pointing
- Selection
- Repetitive commands
- Time-critical actions

Context

Where and under what conditions?

- Driving
- Noisy environment
- Hands occupied
- Accessibility needs
- Mobile/large screen

7 techniques:

- 1) Keyboard
- 2) Mouse
- 3) Foot Mouse
- 4) Eye gaze
- 5) Speech recognition
- 6) ~~Stylus~~ Numeric keyboard
- 7) TV remote

These are a few examples. There can be more.

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Technique	Human Input	Task	Main HCI issues
Keyboard	Hands	Text entry	Speed, accuracy
Mouse	Hands	Pointing & Selection	Target size, distance
Foot Mouse	Foot	Hand's free control	Precision, fatigue
Eye gaze	Eyes	Hand's free selection	Gaze errors, dwell time
Speech Recognition	Voice	Command & dictation	Recognition errors
Keypad	Fingers	Numbers / Pins.	Layout & memorability
TV Remote	Hand	Remote Control	Discoverability & grouping

How to evaluate a particular Technique

1) **Learnability:**

Can a new user understand it quickly?

2) **Efficiency:**

Can experienced Users perform task quickly?

3) **Feedback:**

Does the system clearly show what happened?

4) **Accuracy:**

How often does a user make a mistake?

5) **Accessibility:**

Can users with different abilities use it?

lay/date

6) Context fit:

Does it work well in a real
environment?

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